

Architectural Particular Specification
UNIT MASONRY
SECTION 3.1

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SECTION 3.1

UNIT MASONRY

PART 1 GENERAL

1.01 SUMMARY OF WORKS

- A. The works shall include supply and installation of blockworks and brickworks system including but not limited to the following key elements:-
1. Masonry units and lintels
 2. Mortar and grout
 3. Steel reinforcement and stiffeners
 4. Other associated accessories.

1.02 REFERENCE

Reference specifications and standards latest editions:

- A. BSI – British Standards Institute:
- a. BS 12 - Portland Cement.
 - b. BS 476: Part 20 – Fire Tests on Building Materials and Structures - Method for Determination of the Fire Resistance of Elements of Construction (General Principles).
 - c. BS 476: Part 20 – Fire Tests on Building Materials and Structures - Method for Determination of the Fire Resistance of Elements of Construction (General Principles).
 - d. BS 476: Part 21 – Fire Tests on Building Materials and Structures - Methods for Determination of the Fire Resistance of Loadbearing Elements of Construction.
 - e. BS 729 - Hot Dip Galvanized Coatings on Iron and Steel Articles.
 - f. BS 882 Specification For Aggregates From Natural Sources For Concrete.
 - g. BS 890 Specification For Building Lime.
 - h. BS 4449 - Steel for the reinforcement of concrete — Weldable reinforcing steel — Bar, coil and decoiled product — Specification.
 - i. BS 4482 - Steel wire for the reinforcement of concrete products — Specification.
 - j. BS 4483 Specification For Steel Fabric For The Reinforcement Of Concrete.
 - k. BS 4721 Specification For Ready-Mixed Building Mortars.
 - l. BS 5628 Code Of Practice For Use Of Masonry.
 - m. BS 6073 Specification For Precast Concrete Masonry Units.
 - n. BS 6463 - Quicklime, hydrated lime and natural calcium carbonate.
- B. BS EN - European Standard published by European Committee for Standardization.
- a. BS EN 197: Part 1 - Cement - Composition, specifications and conformity criteria for common cements.
 - b. BS EN 413: Part 1 - Masonry cement - Composition, specifications and conformity criteria.
 - c. BS EN 772 Methods Of Test For Masonry Units.
 - d. BS EN 998: Part 2 - Specification for mortar for masonry - Masonry mortar.
 - f. BS EN 1996 – Eurocode 6 Design of masonry structures.
 - g. BS EN 1996: Part 1: Section 1 - Eurocode 6 Design of masonry structures - General rules for reinforced and unreinforced masonry structures.

- h. BS EN 10088: Part 3 – Technical delivery conditions for semi-finished products, bars, rods, wire, sections and bright products of corrosion resisting steels for general purposes.
 - i. BS EN 10264: Part 4 - Steel Wire and Wire Products Steel Wire for Ropes - Stainless Steel Wire.
 - j. BS EN 12350: Part 2 - Testing Fresh Concrete - Slump Test - Incorporating Corrigendum.
 - k. BS EN 13139 - Aggregates for mortar.
- C. BS ISO - British Standard by International Organization for Standardization:
- a. BS ISO 4998 - Continuous hot-dip zinc-coated carbon steel sheet of structural quality.
 - b. BS ISO 6916: Part 1 - Flexible cellular polymeric materials — Sponge and expanded cellular rubber products — Specification — Sheeting.
 - c. BS ISO 13887 - Cold-reduced steel sheet of higher yield strength with improved formability.
- D. BS EN ISO - European Standard by International Organization for Standardization:
- a. BS EN ISO 1461 - Hot Dip Galvanized Coatings on Fabricated Iron and Steel Articles - Specifications and Test Methods.
 - b. BS EN ISO 10130 - Cold rolled low carbon steel flat products for cold forming Technical delivery conditions.
- E. Local Codes:
- a. The Hong Kong Building (Construction) Regulations.
 - b. The Structural Use of Concrete Hong Kong.
 - c. Hong Kong Code of Practice for Fire Resisting Construction.
 - d. Hong Kong Code of Practice for Wind Effect.
 - e. Hong Kong Construction Standard CS1 for Testing Concrete.
 - f. Hong Kong Construction Standard CS2 for Testing Reinforcement.
 - g. Design Manual - Barrier Free Access 1997

1.03 QUALITY ASSURANCE

(NOT USED)

1.04 PERFORMANCE REQUIREMENTS

A. Loading

Works shall be detailed and installed to achieve adequate and satisfactory performance in accordance with BS EN 1996 (Design of masonry structures) and BS EN 1991-1-4:2005+A1:2010 (Eurocode 1. Actions on structures. General actions. Wind actions) with regard to building use category.

B. Design Standards

1. The design of the works shall be in accordance with the recommendations of BS EN 1996 (Design of masonry structures).
2. The works shall be detailed and installed to allow for and withstand all deflections and tolerances of the main structural elements under all design loads or combination of loads without damage or any reduction in the performance of the works.
3. All fixings shall be capable of providing the correct degree of movement required for the circumstances involved, to suit deflections specified and to provide adequate stability for the works.
4. Movement joints shall be provided to accommodate all movements of the building structure and shall comply with BS EN 1996 (Design of masonry structures).
5. Control joints shall be located at all junctions with adjoining work as indicated on the Design Drawings and in accordance with the recommendations of BS EN 1996 (Design of masonry structures).

C. Unless otherwise specified, all materials shall be of the best quality of the type specified and consistent in colour finish.

1.05 ENVIRONMENTAL REQUIREMENTS

(NOT USED)

1.06 ENVIRONMENTAL, CLIMATIC & PSYCHOMETRIC CONDITIONS

(NOT USED)

1.07 SUBMITTALS

(NOT USED)

1.08 PRE-INSTALLATION CONFERENCE

(NOT USED)

1.09 DELIVERY STORAGE AND HANDLING

- A. Protection of Masonry: During construction, cover tops of walls, projections, and sills with waterproof sheeting at end of each day's work. Cover partially completed masonry when construction is not in progress.
1. Extend cover a minimum of 600mm down both sides of walls and hold cover securely in place.
 2. Where one wythe of multiwythe masonry walls is completed in advance of other wythes, secure cover a minimum of 600mm down face next to unconstructed wythe and hold cover in place.

- B. Stain Prevention: Prevent grout, mortar, and soil from staining the face of masonry to be left exposed or painted. Immediately remove grout, mortar, and soil that come in contact with such masonry.
 - 1. Protect base of walls from rain-splashed mud and from mortar splatter by spreading coverings on ground and over wall surface.
 - 2. Protect sills, ledges, and projections from mortar droppings.
 - 3. Protect surfaces of window and door frames, as well as similar products with painted and integral finishes, from mortar droppings.
 - 4. Turn scaffold boards near the wall on edge at the end of each day to prevent rain from splashing mortar and dirt onto completed masonry.
- C. Hot-Weather Requirements: Comply with hot-weather construction requirements contained in BS EN 1996.
- D. Store masonry units on elevated platforms in a dry location. If units are not stored in an enclosed location, cover tops and sides of stacks with waterproof sheeting, securely tied. If units become wet, do not install until they are dry.
- E. Store cementitious materials on elevated platforms, under cover, and in a dry location. Do not use cementitious materials that have become damp.
- F. Store aggregates where grading and other required characteristics can be maintained and contamination avoided.
- G. Deliver preblended, dry mortar mix in moisture-resistant containers designed for use with dispensing silos. Store preblended, dry mortar mix in delivery containers on elevated platforms, under cover, and in a dry location or in covered weatherproof dispensing silos.
- H. Store masonry accessories, including metal items, to prevent corrosion and accumulation of dirt and oil.

1.10 MOCK-UPS

This part shall be read in conjunction with Clause 8.03 of Specification Preliminaries for the requirements for prototypes and trial areas.

1.11 TESTING

(NOT USED)

1.12 WARRANTY AND MAINTENANCE BOND

This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.

1.13 SPARE PART

This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

PART 2 PRODUCTS

2.01 APPROVED MANUFACTURERS/SUPPLIERS

(NOT USED)

2.02 MATERIALS

A. All block walls shall be built up with concrete or concrete hollow block of not less than 100mm thick unless approved by the Contract Administrator for specific locations. No brick or block wall shall be used as external wall to the building.

B. MASONRY UNITS, GENERAL

1. Defective Units: Referenced masonry unit standards may allow a certain percentage of units to contain chips, cracks, or other defects exceeding limits stated in the standard. Do not use units where such defects will be exposed in the completed Work.
2. Fire-Resistance Ratings: Where indicated on Drawings, provide units that comply with requirements for fire-resistance ratings indicated as determined by testing according to BS 476: Part 20 and 21, by equivalent masonry thickness, or by other means, as acceptable to authorities having jurisdiction.

C. CONCRETE MASONRY UNITS (CMU)

1. Shapes: Provide shapes indicated and as follows, with exposed surfaces matching exposed faces of adjacent units unless otherwise indicated.
2. Provide special shapes for lintels, corners, jambs, sashes, movement joints, headers, bonding, and other special conditions.
3. Provide square-edged units for outside corners unless otherwise indicated.
4. Modular precision units conforming to BS 6073.
5. Provide open-end units at walls to be grouted.
6. Provide closed-end units at walls which will not be grouted.
7. Provide closed units at openings where ends or edges would otherwise be exposed in finish work.
8. Provide special shapes and accessory units at locations indicated on Drawings.
9. Except as specified otherwise, provide units in standard gray color.
10. Faces of exposed units, at guest-visible areas, shall not exhibit visually apparent chips, cracks or similar imperfections when viewed at a distance of 3 m under lighting conditions identical to those of the completed installation.
11. For fire-resistive construction, provide units of minimum equivalent thickness as required by Code or units previously tested and certified for use in appropriate degree of resistive construction.

D. CONCRETE LINTELS

1. Concrete Lintels: BS EN 845: Part 2, matching CMUs in color, texture, and density classification; and with reinforcing bars indicated. Provide lintels with net-area compressive strength not less than CMUs.

E. MORTAR AND GROUT MATERIALS

1. Portland Cement: Comply with BS 12 and BS EN 197. Provide natural color or white cement as appropriate to produce specified mortar color required.

2. Where special requirements are shown on the drawings for other types of cement, they shall conform to British Standard requirements as follows:-
 - a. Rapid Hardening Cement - B.S. 12 / BS EN 197-1
 - b. Portland Blast Furnace Cement - B.S. 146 / BS EN 194-4
 - c. Low Heat Portland Cement - B.S. 1370
 - d. Sulphate Resisting Portland Cement - B.S. 4027
 - e. High Slag Portland Blast Furnace Cement - B.S. 4246 / BS 146

Other types of cement shall only be used with written approval of the Contract Administrator.
3. Hydrated Lime: BS 6463.
4. Masonry Cement: BS EN 413.
5. Mortar Cement: BS EN 197: Part 1 and BS EN 413: Part 1.
6. Colored Cement Product: Packaged blend made from portland cement and hydrated lime, masonry cement, or mortar cement, and mortar pigments, all complying with specified requirements, and containing no other ingredients.
7. Formulate blend as required to produce color indicated or, if not indicated, as selected from manufacturer's standard colors.
8. Pigments shall not exceed 10% of portland cement by weight.
9. Pigments shall not exceed 5% of masonry cement or mortar cement by weight.
10. Aggregate for Mortar: BS EN 13139.
11. For mortar that is exposed to view, use washed aggregate consisting of natural sand or crushed stone.
12. For joints less than 6mm thick, use aggregate graded with 100 percent passing the 1.18mm sieve.
13. White-Mortar Aggregates: Natural white sand or crushed white stone.
14. Colored-Mortar Aggregates: Natural sand or crushed stone of color necessary to produce required mortar color.
15. Mortar color pigments: Pure natural or synthetic iron oxides; unless selected otherwise by Owner, color to closely match CMUs.
16. Aggregate for Grout: BS EN 13139.
17. Water: Potable.
18. All cement to be used on to the site or elsewhere for the works shall be delivered in the original sealed and branded bags of the manufacturer or in bulk containers endorsed by the Architect.
19. Bagged cement shall not be stacked up more than 15 bags high in order to prevent hard lumps being formed.
20. Sand shall be thoroughly washed with no dirt, shells and other contaminants before use. Refer also ASD GS 2012.
21. Manufactured sand is to be of a type suitable for general use in mortars and plasters, and to conform to the grading limits set out below.
22. Sand shall be free from any deleterious materials such as clay, flaky or elongated particles, mica, shale or other laminated materials, organic impurities, iron pyrites. Sand shall not contain more than 1% of soluble sulphate salts. In general, and except as specified herein, the sand is to comply with B.S. 1199 and 1200 / BS EN 13139.

23. Sand for mixed not incorporating lime shall be crushed granite or other durable rock fines or natural sand free from salt.
24. Do not use admixtures in mortar unless specified or approved in writing by the Contract Administrator.
25. Do not use calcium chloride or any admixtures containing calcium chloride.
26. Pigments for colouring mortar shall be an approved proprietary brand to B.S. 1014 / BS EN 12878.

F. REINFORCEMENT

1. The Contractor shall include reinforcement in the brickwork / blockwork.
2. Brick / block reinforcement shall conform with B.S. 405 and be made from galvanized steel reinforcement strip mesh not less than 0.5mm thick and shall be well lapped at angles and passings.
3. Brick / block reinforcement shall be laid on an even bed of mortar in a continuous strip with 225mm laps at joints and full laps at angles.
4. Keep back 20mm from face of external work, 12mm back from face of internal work, and finish mortar joint to normal thickness.
5. Lay on layers at not more than 1200mm centres vertically.
6. Epoxy-coated Steel Reinforcing Bars: BS 4449.
7. Masonry Joint Reinforcement, General: BS EN 845: Part 3.
8. Interior Walls: Hot-dip galvanized, carbon steel.
9. Exterior Walls: Stainless steel.
10. Wire Size for Side Rods: 3.77mm to 4.76mm diameter.
11. Wire Size for Cross Rods: 3.77mm to 4.76mm diameter.
12. Spacing of Cross Rods, Tabs, and Cross Ties: Not more than 407mm at centers.
13. Provide in lengths of not less than 3m, with prefabricated corner and tee units.
14. Masonry Joint Reinforcement for Single-Wythe Masonry: Either ladder or truss type with single pair of side rods

G. TIES AND ANCHORS

1. All external and internal wall panels shall be tied to concrete columns, or posts with fishtail anchors in the dovetail slots cast in the concrete. The anchors or ties shall be provided in every third course to external walls and every sixth course to internal walls.
2. External walls shall be tied to concrete columns or posts by galvanised steel ties.
3. Internal walls shall be tied to the concrete columns, or posts with galvanised steel ties built into the wall. All internal walls shall be restrained on the top with galvanised mild steel angles. For non-load bearing walls, the head fixing of the wall should have a minimum 20mm gap to allow for differential movement of opening or closing of 20mm min. The head fixing / ties are to allow for this vertical differential movement. Galvanised dovetailed masonry slots / anchors and galvanised steel straps for internal walls shall be provided at 1200mm centres or as otherwise shown on the drawings.
4. Materials: Provide ties and anchors specified in this article that are made from materials that comply with the following unless otherwise indicated.

- a. Hot-Dip Galvanized, Carbon-Steel Wire: BS 4482 wire with BS 729 and BS EN ISO 1461 coating.
 - a. Stainless-Steel Wire: BS EN 10264: Part 4, grade 1.4401 with BS EN 10088: Part 3 coating.
 - b. Steel Sheet, Galvanized after Fabrication: BS ISO 13887 Commercial Steel, with BS 729 and BS EN ISO 1461 coating.
 - c. Stainless-Steel Sheet: BS EN ISO 10130, grade 1.4401.
 - d. Steel Plates, Shapes, and Bars: BS ISO 4998.
5. Corrugated Metal Ties: Metal strips not less than 22mm wide with corrugations having a wavelength of 7.6 to 12.7 mm and an amplitude of 1.5mm to 2.5mm made from the following:
 - a. Interior Walls: 0.76mm thick, steel sheet, galvanized after fabrication.
 - e. Exterior Walls: 0.79mm thick, stainless-steel sheet.
6. Wire Ties, General: Unless otherwise indicated, size wire ties to extend at least halfway through veneer but with at least 16mm cover on outside face. Outer ends of wires are bent 90 degrees and extend 50mm parallel to face of veneer.
7. Adjustable Anchors for Connecting to Structural Steel Framing: Provide anchors that allow vertical or horizontal adjustment but resist tension and compression forces perpendicular to plane of wall.
 - a. Anchor Section for Welding to Steel Frame:
 - Interior Walls: Crimped 6.35mm diameter, hot-dip galvanized steel wire.
 - Exterior Walls: Crimped 6.35mm diameter, stainless-steel wire.
 - f. Tie Section: Triangular-shaped wire tie, sized to extend within 25 mm of masonry face, made from the following:
 - Interior Walls: Crimped 4.76mm diameter, hot-dip galvanized steel wire.
 - Exterior Walls: Crimped 4.76mm diameter, stainless-steel wire.
8. Adjustable Anchors for Connecting to Concrete: Provide anchors that allow vertical or horizontal adjustment but resist tension and compression forces perpendicular to plane of wall.
 - a. Corrugated Metal Ties: Metal strips not less than 22mm wide with corrugations having a wavelength of 7.6mm to 12.7mm and an amplitude of 1.5mm to 2.5mm) made from the following with dovetail tabs for inserting into dovetail slots in concrete and sized to extend to within 25mm of masonry face.
 - Interior Walls: **2.01mm** thick steel sheet, galvanized after fabrication.
 - Exterior Walls: **1.59mm** thick stainless-steel sheet.
9. Partition Top Anchors: 2.66mm thick metal plate with 9.5mm diameter metal rod 152mm long welded to plate and with closed-end plastic tube fitted over rod that allows rod to move in and out of tube. Fabricate from the following:
 - Interior Walls: Steel, galvanized after fabrication.
 - Exterior Walls: Stainless-steel sheet.
10. Rigid Anchors: Fabricate from steel bars 38 mm wide by 6.35mm thick by 610 mm long, with ends turned up 51mm or with cross pins unless otherwise indicated.

- a. Corrosion Protection: Epoxy coating 0.51mm thick.
- 11. Head Restraint: Vertically sliding angles assembly with hole at upper angle for fixing overhead masonry unit being supported:
 - a. Material: Stainless steel, grade 1.4301.
 - g. Height: Adjustable for 100mm to 140mm high masonry unit.
- 12. Wall Tie and Debonding Sleeve.
 - a. Wall Tie:
 - General: Metal strip with hole at side for fixing side masonry unit being supported.
 - Material: Stainless steel, grade 1.4301.
 - Plain ended strip.
Size: 250mm long.
 - Plain ended horizontal angle.
Size: 125mm long.
 - h. Debonding Sleeve: Hollow steel component for allowance of movement and tolerance of masonry unit by covering wall tie.
 - Material: Stainless steel, grade 1.4301.
 - Size: 120mm long
- 13. Wind Post: Storey-height metal C-shaped channel to provide stability of masonry structure.

2.03 TYPICAL STEEL STIFFENERS SCHEDULE FOR BRICK / BLOCK WALL PARTITION

For all block / brick walls to be constructed, appropriate steel stiffeners / reinforcements are to be incorporated as per the following schedule. Calculations shall be prepared by the Contractor's Engineer and verified by the Contractor's Register Structural Engineer confirming that the Block Wall systems have been designed to satisfy the specified performance requirements, in particular regard to the specified loading criteria and maximum deflection limits. Shop-drawings / structural calculations should also be submitted for LAC/ consultants' confirmation of no adverse comment before execution on site. The table below is provided for information only.

At the top of the vertical steel stiffener, provide an allowance for plus or minus 20mm min vertical movement.

Wall Thickness	H (m) Wall Full Height	h(m) Height between horizontal steel stiffeners	L(m) Width between vertical steel stiffeners	Steel Stiffeners	
100	3 ~ 6	3	2.5	Vertical: Horizontal:	TWIN 102 x 51 channel 102 x 51 channel
150	3 ~ 6	3	3.5	Vertical: Horizontal:	TWIN 127 x 64 channel 102 x 51 channel
150	6 ~ 9	6	4	Vertical: Horizontal:	TWIN 152 x 152 x 37 UC 152 x 76 channel
150	9 ~ 12	6	4	Vertical: Horizontal:	203 x 203 x 52 UC 152x 76 channel
200	6 ~ 10	6	6	Vertical: Horizontal:	203 x 203 x 46UC 203 x 76 channel
	10 ~ 12	6	6	Vertical: Horizontal:	203 x 203 x 71UC 203 x 76 channel

2.04 MISCELLANEOUS MASONRY ACCESSORIES

- A. Compressible Filler: Premolded filler strips complying with BS ISO 6916: Part 1; compressible up to 35%; of width and thickness indicated; formulated from neoprene, urethane, or PVC.
- B. Bond-Breaker Strips: Asphalt-saturated, organic roofing felt.
- C. Weep/Vent Products: Use the following unless otherwise indicated:

1. Cellular Plastic Weep/Vent: One-piece, flexible extrusion made from UV-resistant polypropylene copolymer, full height and width of head joint and depth 3mm less than depth of outer wythe, in color selected from manufacturer's standard.
- D. Reinforcing Bar Positioners: Wire units designed to fit into mortar bed joints spanning masonry unit cells and hold reinforcing bars in center of cells. Units are formed from 3.77mm steel wire, hot-dip galvanized after fabrication. Provide units designed for number of bars indicated.

2.05 MORTAR AND GROUT MIXES

- A. Mortar: Mix proportions as approved by testing laboratory
1. Portland cement
 2. Hydrated lime
 3. Mortar sand
 4. Water
 5. Color pigments
- B. Grout: Mix proportions as approved by testing laboratory
1. Portland cement
 2. Grout sand
 3. Pea gravel
 4. Sika Grout Aid by Sika or WRDA with HYCOL by Grace or equivalent
- C. Measurements: Proportion by accurate volume measurements. Measure in suitable calibrated devices that can be easily and accurately checked at any time.
1. Add water for workable consistency
 2. Shovel measurements not permitted.
- D. Mixing mortar:
1. With lime: Place sand, cement, and water in mixer in order, while mixer is running. Mix for 2 min., add lime and Sika-1 admixture, and continue mixing until a uniform mass is secured, but in no case less than 10 min.
 2. Retempering time limit: Retemper on mortar boards, when required, by adding water into a basin formed by mortar, and working mortar into it. Do not dash or pour water over mortar.
 - a. Do not retemper mortar which has set, or has become harsh or non-plastic
 - i. Discard mortar which has not been used within 1 hr. after original mixing.
- E. Preblended, packaged mortar: Use preblended, packaged mortar mixes conforming to specified compressive strength and material requirements.
- F. Mixing grout: Mix the cement, sand, gravel, water with admixture in accord with the manufacturer's recommendation.
- G. Ready-mix grout: Grout batched off-site and delivered by mixer truck shall be subject to the same procedures and controls prescribed in GENERAL SPECIFICATION
- H. Equipment for mixing and handling mortar and grout shall be acceptable to Owner.
- I. Do not use batches of less than one sack of cement or fractional sack batches.

PART 3 EXECUTION

3.01 PREPARATION

- A. Examine conditions, with installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.
- B. Verify that reinforcing dowels are properly placed.
- C. Before installation, examine rough-in and built-in construction for piping systems to verify actual locations of piping connections.
- D. Proceed with installation only after unsatisfactory conditions have been corrected.

3.02 INSTALLATION, GENERAL

- A. Thickness: Build cavity and composite walls and other masonry construction to full thickness shown. Build single-wythe walls to actual widths of masonry units, using units of widths indicated.
- B. Build chases and recesses to accommodate items specified in this and other Sections.
- C. Leave openings for equipment to be installed before completing masonry. After installing equipment, complete masonry to match the construction immediately adjacent to opening.
- D. Use full-size units without cutting if possible. If cutting is required to provide a continuous pattern or to fit adjoining construction, cut units with motor-driven saws; provide clean, sharp, unchipped edges. Allow units to dry before laying unless wetting of units is specified. Install cut units with cut surfaces and, where possible, cut edges concealed.
- E. Select and arrange units for exposed unit masonry to produce a uniform blend of colors and textures.
 - 1. May mix units from several pallets or cubes as they are placed.

3.03 TOLERANCES

- A. Dimensions and Locations of Elements:
 - 1. For dimensions in cross section or elevation do not vary by more than plus 12 mm or minus 6 mm.
 - 2. For location of elements in plan do not vary from that indicated by more than plus or minus 12 mm.
 - 3. For location of elements in elevation do not vary from that indicated by more than plus or minus 6 mm in a story height or 12 mm total.
- B. Lines and Levels:
 - 1. For bed joints and top surfaces of bearing walls do not vary from level by more than 6 mm in 3 m, or 12 mm maximum.
 - 2. For conspicuous horizontal lines, such as lintels, sills, parapets, and reveals, do not vary from level by more than 3 mm in 3 m, 6 mm in 6 m, or 12 mm maximum.
 - 3. For vertical lines and surfaces do not vary from plumb by more than 6 mm in 3 m, 9 mm in 6 m, or 12 mm maximum.

4. For conspicuous vertical lines, such as external corners, door jambs, reveals, and expansion and control joints, do not vary from plumb by more than 3 mm in 3 m, 6 mm in 6 m, or 12 mm maximum.
5. For lines and surfaces do not vary from straight by more than 6 mm in 3 m, 9 mm in 6 m, or 12 mm maximum.
6. For vertical alignment of exposed head joints, do not vary from plumb by more than 6 mm in 3 m, or 12 mm maximum.
7. For faces of adjacent exposed masonry units, do not vary from flush alignment by more than 1.5 mm except due to warpage of masonry units within tolerances specified for warpage of units.

C. Joints:

1. For bed joints, do not vary from thickness indicated by more than plus or minus 3 mm, with a maximum thickness limited to 12 mm.
2. For exposed bed joints, do not vary from bed-joint thickness of adjacent courses by more than 3 mm.
3. For head and collar joints, do not vary from thickness indicated by more than plus 9 mm or minus 6 mm.
4. For exposed head joints, do not vary from thickness indicated by more than plus or minus 3 mm. Do not vary from adjacent bed-joint and head-joint thicknesses by more than 3 mm.
5. For exposed bed joints and head joints of stacked bond, do not vary from a straight line by more than 1.5 mm from one masonry unit to the next.

3.04 INSTALLATION - REINFORCEMENT

A. Vertical bars:

1. Low-lift grouting:
 - a. Placed prior to CMUs.
 - j. Tied to foundation dowels.
 - k. Braced at 1200 mm intervals.
 2. High-lift grouting
 - a. Place after masonry is laid, prior to grouting.
 - l. Support at 1200 mm intervals with steel centering devices laid in bed joints.
- B. Horizontal bars laid in place as work progresses.
- C. Ladder/truss type reinforcing:
1. Unless indicated otherwise on Drawings, place reinforcing in first and second bed joints above lintel and below sill of all openings. At second bed joint above and below opening, extend reinforcing 600 mm beyond joint at each side of opening.
 2. Reinforce all other wall areas at 400 mm o.c. vertically.
 3. Reinforce all corners and tees (wall intersections).
 4. Lap side wires 150 mm at splices.
 5. Place reinforcing to obtain 15 mm minimum cover at wall face.

3.05 INSTALLATION – CONCRETE MASONRY UNITS

- A. Workmanship: Except as specified otherwise, conform to BS 6073.

1. Erect masonry plumb and true to line.
2. Keep masonry clean during and after laying.
3. Lay out and incorporate all embedded items.
4. Assist other trades with built-in items which require cutting and fitting of masonry.
5. Cut block units with a steel saw or Carborundum wheel. Do not trowel or chisel cut.

B. Shoring:

1. Provide, in place, temporary shoring for lintels, strong enough to carry load without deflecting.
2. Remove temporary shoring after masonry has been in place 28 days.

C. Laying block: Remove all dirt and dust from surfaces before laying.

1. Foundation preparation: Sandblast tops of concrete starting surfaces; wash off with high-pressure water jet. Slurry coat surfaces with a pre-proportioned setting grout (Master Builder's Emaco T415 Mortar or Owner-approved equivalent) mixed to proper consistency.
2. After bond bed has hardened slightly, spread mortar to required joint thickness. Lay blocks with 10 mm mortar bed on entire horizontal surface. Fill head joints solid, shove tightly to adjoining units. All joints shall be 10 mm.
 - a. Hold racking to a minimum.
 - m. No toothing shall be allowed.
 - n. If it becomes necessary to move a unit after it has been set in place, remove unit, discard mortar, and reset unit in fresh mortar.
3. Anchor bolts: Provide 25 mm minimum grout space around all protruding bolts.
4. Bond: Unless indicated otherwise on Drawings, lay all units in common running bond.
5. Finish joint treatment:
 - a. At surfaces to receive a subsequently-applied waterproof finish (e.g., membrane waterproofing, or high-build coatings), cut joints flush; do not tool.
 - o. At exposed surfaces (exterior and interior), cut joints flush and tool concave to a dense, uniform surface.
6. Grouting: Unless indicated otherwise on Drawings, completely fill all cells with grout.
7. Masonry fill insulation: Fill all cells in exterior walls enclosing interior space which are not filled solid with grout, solid with insulation.

D. Hollow metal frames:

1. Locate door frames accurately, erect plumb, securely attach to floor, and brace in position prior to start of masonry construction.
 - a. Furnish frames with adjustable anchors. Install anchors at horizontal masonry joints.
 - p. Fill interior of frames solid with mortar or grout as walls are constructed.
2. Provide temporary wood spreaders from jamb to jamb and from head to floor to ensure that jambs do not bow-in or distort from a straight line, or deflect from superimposed loads during construction.

3.06 LOW-LIFT GROUTING

- A. After mortar in joints has firmly set, cores cleaned of mortar and debris, reinforcing is properly in place and inspected, grout cells in 1200 mm maximum lifts, using specified pea gravel grout mix.
- B. Grout walls solid, without voids.
- C. Grout may be placed by pump, tremie, or bucket, using hoppers to avoid spilling on exposed surfaces.
- D. Place an initial 600 mm high lift all around, thoroughly compact, then place balance of each lift, compacting again through the total lift using hardwood spading sticks or pencil vibrators.
- E. Stop grout pours 40 mm below top of CMU at each lift.
- F. Remove and discard spilled grout from upper units before grout hardens.
- G. Bracing: Adequately brace walls against wind and other forces during construction.

3.07 HIGH-LIFT GROUTING OPTION

- A. Contractor may utilize high-lift grouting method, provided the following qualifications and requirements are met. High-lift grouting applies only to cell sizes available with 200 mm and wider block units.
- B. Contractor qualification: As specified.
- C. Contractor shall be responsible for all blowouts or displacement of CMUs because of hydrostatic pressures, voids, or ungrouted areas due to bridging of grout or blocking of cells from any cause.
- D. Use bond beam units, inverted for starter course, and omit alternate blocks or cut openings in alternate face shell on bottom course for clean-outs.
- E. Use a hardwood stick to knock off projecting mortar fins. Wash out every cell thoroughly using a water jet which has sufficient force to remove all mortar from interior of cells and from reinforcing steel.
- F. Plug each clean-out by setting a "soap" in mortar into the opening and securely bracing it in place to prevent displacement. If masonry is not exposed in finish work, clean-outs may be formed.
- G. Grouting:
 - 1. Grout masonry cells solid, without voids.
 - 2. Do not place grout until masonry has set a minimum of 24 hr.
 - 3. Pump grout from mixer into grout cell space as rapidly as practical. Discard grout not in place within 1-1/2 hr. after water was first added to batch.
 - 4. Place grout with maximum slump without segregation. Place in a continuous pour, in maximum lifts of 900 mm, with approximately 20 min. elapsed time between any two successive lifts.
- H. Compacting:
 - 1. Compact and recompact grout using 200 mm light weight flexible cable vibrators.
 - 2. Compact completely to bottom of first lift immediately after placement, and in the case of subsequent lifts, through previously-placed lift.

3. Recompact top lift not sooner than 30 min. after grout has been placed.
4. Do not permit vibrating of reinforcing steel.

I. Bracing: Adequately brace walls against wind and other forces during construction.

3.08 ANCHORING MASONRY TO STRUCTURAL STEEL AND CONCRETE

- A. Anchor masonry to structural steel and concrete where masonry abuts or faces structural steel or concrete to comply with the following:
1. Provide an open space not less than 25mm wide between masonry and structural steel or concrete unless otherwise indicated. Keep open space free of mortar and other rigid materials.
 2. Anchor masonry with anchors embedded in masonry joints and attached to structure.
 3. Space anchors as indicated, but not more than 610mm at centers vertically and 915 mm at centers horizontally.

3.09 CONTROL AND EXPANSION JOINTS

- A. General: Install control and expansion joint materials in unit masonry as masonry progresses. Do not allow materials to span control and expansion joints without provision to allow for in-plane wall or partition movement.
- B. Form control joints in concrete masonry as follows:
1. Location: As indicated on Drawings.
 2. Method: As specified in Division 07 Section "Expansion Control".
- C. Provide horizontal, pressure-relieving joints by either leaving an air space or inserting a compressible filler of width required for installing sealant and backer rod specified in Division 07 Section "Joint Sealants," but not less than 10mm.

3.10 LINTELS

- A. Install steel lintels where indicated.
- B. Provide concrete lintels where openings of more than 610mm wide are shown without structural steel or other supporting lintels.
- C. Provide minimum bearing of 200mm at each jamb unless otherwise indicated.

3.11 WEEP HOLES AND VENTS

- A. General:
1. Install weep holes in masonry at shelf angles, lintels, ledges, other obstructions to downward flow of water in wall, and where indicated on Drawings.
 2. Install vents at shelf angles, ledges, and other obstructions to upward flow of air in cavities, and where indicated on Drawings.
- B. Install weep as follows:
1. Use proprietary weep/vent products or open head joints to form weep holes.
 2. Space weep holes 600mm at centers unless otherwise indicated.
 3. Cover cavity side of weep holes with plastic insect screening at cavities insulated with loose-fill insulation.

- C. Install vents in location indicated on Drawings. Use proprietary weep/vent products or open head joints to form vents.

3.12 CURING, CLEANING AND PROTECTION

- A. Remove and replace masonry units that are loose, chipped, broken, stained, or otherwise damaged or that do not match adjoining units. Install new units to match adjoining units; install in fresh mortar, pointed to eliminate evidence of replacement.
- B. Pointing: During the tooling of joints, enlarge voids and holes, except weep holes, and completely fill with mortar. Point up joints, including corners, openings, and adjacent construction, to provide a neat, uniform appearance. Prepare joints for sealant application, where indicated.
- C. In-Progress Cleaning: Clean unit masonry as work progresses by dry brushing to remove mortar fins and smears before tooling joints.
- D. Final Cleaning: After mortar is thoroughly set and cured, clean exposed masonry as follows:
 - 1. Remove large mortar particles by hand with wooden paddles and nonmetallic scrape hoes or chisels.
 - 2. Test cleaning methods on sample wall panel; leave one-half of panel uncleaned for comparison purposes.
 - 3. Protect adjacent stone and non-masonry surfaces from contact with cleaner by covering them with liquid strippable masking agent or polyethylene film and waterproof masking tape.
 - 4. Wet wall surfaces with water before applying cleaners; remove cleaners promptly by rinsing surfaces thoroughly with clear water.
 - 5. Clean masonry with a proprietary acidic cleaner applied according to manufacturer's written instructions.
 - 6. Clean concrete masonry by cleaning method appropriate for the type of stain on exposed surfaces.
- E. Do not saturate masonry with water for curing and other purposes.
- F. Where atmosphere is dry, dampen wall surface with a very light fog spray for 3 days to help cure mortar in joints.

END OF SECTION